

Data Exploration and Preparation

Diploma in DAB – L7

CCT College Dublin

Exploratory Data Analysis

Week 4

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Outline

- Exploratory Data Analysis (EDA)
- Purpose of EDA
- Data Exploration Methods
- Data Quality and Preparation
- Data Science Project: Planning
- CRISP-DM and SAMMA

Exploratory Data Analysis (EDA)

- EDA is the process of examining and investigating a dataset before formal modelling.
- It helps us understand the **structure, quality and characteristics** of the data.
- EDA is based on
  - Descriptive statistics
  - Data visualisation
  - Data quality checking
  - Identification of patterns and relationships
- The purpose is to move from **raw data → useful information → informed decisions**.
- EDA is closely connected to data understanding and data preparation.
- **Question:** *What is the data telling us, and how can we use that information?*

Purpose of EDA

- Before analysing or modelling data, we need to understand the following
- EDA can help us to identify
    - Missing and Incorrect or inconsistent values
    - Duplicate observations
    - Unusual observations or outliers
    - Distribution of variables
    - Relationships between variables
- For example, in a student dataset, we might ask
    - Are enrolments increasing in the courses of not?
    - Which programmes have the most students?
    - Is student satisfaction changing?
- CRISP-DM guidance places exploration of data quality, tables and graphics within the **Data Understanding** phase.

Data Exploration Methods

- **EDA** normally combines **numerical summaries and visualisation**.
- **Numerical techniques:** Mean, Median, Mode, Minimum and maximum, Standard deviation, Frequency counts and Correlation
- **Visual Techniques:**
- Histograms → examine distributions
- Box plots → identify spread and possible outliers
- Bar charts → compare categories
- Scatter plots → investigate relationships
- Line charts → examine trends over time
- Pair plots → compare several numerical variables
- A graph should help to answer a question rather than simply make the analysis.

Data Quality and Preparation

- The data analysis depends on good-quality data. Common data-quality issues include
  - Missing values
  - Duplicate records
  - Incorrect data types
  - Inconsistent categories
  - Outliers
  - Invalid values
  - Different measurement formats

```
# Mean imputation
df["Average_Mark"].fillna(
    df["Average_Mark"].mean()
)
# Median imputation
df["Student_Satisfaction"].fillna(
    df["Student_Satisfaction"].median()
)
# Mode imputation
df["Study_Mode"].fillna(
    df["Study_Mode"].mode()[0]
)
```

- **Examples of Preparation:**
- The choice of method should be justified, because changing or imputing data can affect later analysis. IBM notes that data quality can influence model quality and identifies data assessment and preparation as important early stages of CRISP-DM.

Data Science Project

Planning

- **EDA** should not be performed without a clear objective. A structured project asks

1. **What is the problem?**
   Example: Understand changes in the dataset.
2. **What data do we need?**
   For example: year, programme, study mode and enrolment.
3. **Is the data suitable?**
   Check completeness, accuracy and consistency.
4. **What analysis should we perform?**
   Use appropriate statistics and visualisations.
5. **How will we validate our findings?**
   Check calculations, assumptions and reproducibility.
6. **How will findings be communicated?**
   Use appropriate charts, tables and explanations.

- Two useful frameworks for structuring analytical work are **CRISP-DM** and **SEMMA**.

CRISP-DM

- **CRISP-DM = Cross-Industry Standard Process for Data Mining**
- It contains six phases

1. **Business Understanding:** Define the problem, objectives and success criteria.
2. **Data Understanding:** Collect, describe, explore and assess the data.
3. **Data Preparation:** Clean, transform and organise the data.
4. **Modelling:** Select and apply appropriate analytical/model techniques.
5. **Evaluation:** Determine whether the results meet the original objectives.
6. **Deployment:** Communicate or implement the results.

- CRISP-DM is **iterative rather than strictly linear**. A discovery during modelling, for example, may require returning to data preparation. IBM

CRISP-DM

Example

- Imagine that we want to investigate student performance and enrolment.
- **Business Understanding**
  Question: How are enrolment, performance and satisfaction changing?
- **Data Understanding**
  Examine variables such as Year, Programme, Study Mode, Enrolments, Graduates, Average Mark and Satisfaction.
- **Data Preparation**
  Handle missing values, duplicates and incorrect data types.
- **Modelling / Analysis**
  Examine relationships and, where appropriate, develop statistical or predictive models.
- **Evaluation**
  Check whether the analysis answers the original questions and whether the results are reliable.
- **Deployment**
  Present findings using a report, dashboard or presentation.
- This illustrates why **EDA is not an isolated task**; it contributes to a wider analytical process.

SEMMA

- **SEMMA** is associated with the SAS data-mining process. It stands for
- **S – Sample:** Select data that adequately represents the problem.
- **E – Explore:** Use statistics and visualisations to investigate patterns, trends and anomalies.
- **M – Modify:** Clean, transform, select and create variables.
- **M – Model:** Apply appropriate analytical or predictive models.
- **A – Assess:** Evaluate and compare the results.
- SAS documentation specifically includes tasks, such as replacing missing values and transforming variables within the **Modify** stage.
- **Source:** SAS Enterprise Miner documentation. SAS Support

CRISP-DM and SEMMA

Comparison

- Both approaches encourage a structured and iterative analytical process, but their emphasis differs.
- CRISP-DM explicitly begins with the wider business/problem objective and ends with deployment. SEMMA describes a process centered more directly on preparing, exploring and modelling data.
- IBM's CRISP-DM documentation explicitly covers six phases from requirements to deployment, whereas SAS defines its five SEMMA stages around the data-mining process. IBM.
- **Discussion Question:**
  Which framework would be more appropriate for your current data analytics project, and why?

CRISP-DM	SEMMA
Business Understanding	—
Data Understanding	Sample / Explore
Data Preparation	Modify
Modelling	Model
Evaluation	Assess
Deployment	—

Resources/ References

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